Bonding worksheet

1. **Which substance is likely to have a metallic structure**?

a) melts at 2800ºC, insoluble in water, non-conductor of electricity when molten or solid
b) melts at 2600ºC, dissolves in water, non-conductor of electricity when solid, undergoes electrolysis in aqueous solution
c) melts at 17ºC, insoluble in water, non-conductor of electricity either when solid or molten
d) melts at 1400ºC, insoluble in water, good conductor of electricity either when solid or molten

2. Sulphur is a solid non-metallic element at room temperature, so it is?

a) a substance with a high melting point
b) easily bent into shape
c) a good conductor of electricity
d) a poor conductor of heat

3. Copper is a metallic element so it is likely to be a?

a) substance with a low melting point
b) substance with a shiny surface
c) poor conductor of heat
d) poor conductor of electricity

4. Water is a covalent compound. When this compound is formed from hydrogen and oxygen?

a) electrons are shared
b) protons are transferred
c) electrons are transferred
d) protons are shared

5. From four substances described below, which is most likely to consist of a giant covalent structure?

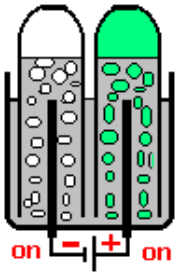
a) melting point 20ºC, molten substance doesn't conduct electricity
b) melting point 1500ºC, molten substance is poor electrical conductor
c) melting point 480ºC, molten substance readily conducts electricity
d) melting point 1150ºC, molten substance readily conducts electricity

6. Aluminium fluoride is an ionic compound. **When this compound is formed from Aluminium and fluorine**?

a) electrons are transferred
b) electrons are shared
c) protons are transferred
d) protons are shared

7. Copper is a metallic element so it is likely to be a?

a) substance easily hammered into shape
b) poor conductor of heat
c) substance with a low melting point
d) poor conductor of electricity



8. **Molten or dissolved sodium chloride conducts electricity. The current is carried in the melt by?**

a) metal atoms
b) ions
c) molecules
d) electrons

9. When an ionic bond is formed between atoms of different elements?

a) electrons are transferred
b) electrons are shared
c) protons are transferred
d) protons are shared

10. Sodium chloride is a typical ionic compound formed by combining a metal with a non-metal. Sodium chloride will?

a) not conduct electricity when dissolved in water
b) not conduct electricity when solid
c) have a low melting point
d) consist of NaCl molecules